Binomial thinning credibility model for total and extreme claim counts

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Abstract

To be written.

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1. Introduction

In many insurance portfolios, two claim counts are recorded for the same risk unit and period: the total number of claims and the number of extreme claims among them, for example the claims whose amounts exceed a large-loss threshold. The two counts are nested, since the number of extreme claims cannot exceed the total number of claims, and they are observed repeatedly over time. A joint model for such data should respect the nesting, accommodate time-varying covariates, and allow the past experience of a risk unit to update the predictions of both counts.

This paper proposes a model in which the extreme claim count is a binomial thinning of the total claim count. The total claim count of a region follows a Poisson distribution with a gamma random effect, and each claim is classified as extreme with a probit probability that depends on a Gaussian random effect. Under independent and static random effects, we show that the likelihood factorizes into a Poisson-gamma part and a Bernoulli-probit part, so that the two sets of parameters can be estimated separately. We further derive the posterior distributions of the random effects and the exact Bayesian one-step predictions of both counts in closed form.

The paper is organized as follows. Section 2 introduces the notation and two auxiliary results on Gaussian random effects in probit models. Section 3 presents the model and the main results.

2. Preliminaries

2.1. Notation

Let $\mathbb{N}$ denote the set of positive integers and $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. We write $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{R}_{>0}$ for the real line, the nonnegative real line, and the positive real line, respectively. For $k \in \mathbb{N}$, vectors in $\mathbb{R}^k$ are column vectors, and we write $x_{1:k} := (x_1, \ldots, x_k)^\top$. The $k \times k$ identity matrix is denoted by $\boldsymbol{I}_k$. Inequalities between vectors are understood componentwise. A product over an empty index set equals 1, and a sum over an empty index set equals 0.

We use the following distributions.

- Gamma(a, b): gamma distribution with shape $a \in \mathbb{R}_{>0}$ and rate $b \in \mathbb{R}_{>0}$, so that its mean is a/b.
- N(μ, σ^2): normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$.
- MVN$_k(\mu_{1:k}, \boldsymbol{\Sigma})$: k-variate normal distribution with mean $\mu_{1:k} \in \mathbb{R}^k$ and positive semidefinite covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$.
- Pois(λ): Poisson distribution with mean $\lambda \in \mathbb{R}_{>0}$.
- Bin(n, p): binomial distribution with $n \in \mathbb{N}_0$ trials and success probability $p \in (0, 1)$, where Bin$(0, p)$ is the point mass at 0.

Let ϕ and Φ denote the density and the distribution function of N$(0, 1)$. For a positive semidefinite matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$, let

$$\Phi_k(x_{1:k}; \boldsymbol{\Sigma}) := \mathbb{P}\left(X_{1:k} \leq x_{1:k}\right), \qquad X_{1:k} \sim \text{MVN}_k(0_{1:k}, \boldsymbol{\Sigma}), \quad x_{1:k} \in \mathbb{R}^k,$$

where $0_{1:k} \in \mathbb{R}^k$ is the zero vector. For $k = 0$, we set $\Phi_0 := 1$.

2.2. Gaussian random effects in probit models

The following identity converts the expectation of a product of probit functions of a Gaussian vector into a multivariate normal distribution function.

Lemma 1 (Gaussian product-probit identity)**.** *Let $X_{1:k} \sim \mathrm{MVN}_k(\mu_{1:k}, \boldsymbol{\Sigma})$, where $\mu_{1:k} \in \mathbb{R}^k$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{k\times k}$ is positive semidefinite. Then*

$$\mathbb{E}\left[\prod_{j=1}^{k} \Phi(X_j)\right] = \Phi_k\left(\mu_{1:k}; \boldsymbol{I}_k + \boldsymbol{\Sigma}\right).$$

Proof. Let $W_{1:k} \sim \mathrm{MVN}_k(0_{1:k}, \boldsymbol{I}_k)$ be independent of $X_{1:k}$. Since $W_1, \ldots, W_k$ are independent standard normal random variables,

$$\prod_{j=1}^{k} \Phi(X_j) = \mathbb{P}\left(W_{1:k} \le X_{1:k} \mid X_{1:k}\right).$$

Taking expectations gives

$$\mathbb{E}\left[\prod_{j=1}^{k} \Phi(X_j)\right] = \mathbb{P}\left(W_{1:k} \le X_{1:k}\right) = \mathbb{P}\left(W_{1:k} - (X_{1:k} - \mu_{1:k}) \le \mu_{1:k}\right).$$

Since $W_{1:k}$ and $X_{1:k}$ are independent Gaussian vectors,

$$W_{1:k} - (X_{1:k} - \mu_{1:k}) \sim \mathrm{MVN}_k\left(0_{1:k}, \boldsymbol{I}_k + \boldsymbol{\Sigma}\right),$$

which proves the result. □

Lemma 1 does not require $\boldsymbol{\Sigma}$ to be positive definite. This is essential for the next result, in which a single scalar random effect is shared by all components, so that the covariance matrix has rank one.

Lemma 2 (Bernoulli-probit random effect)**.** *Let $\Theta \sim \mathrm{N}(0, \sigma^2)$ with $\sigma > 0$, and let $k \in \mathbb{N}$, $u_{1:k} \in \{-1, 1\}^k$, and $h_{1:k} \in \mathbb{R}^k$. Then*

$$\begin{aligned}\mathbb{E}\left[\prod_{j=1}^{k} \Phi\left(u_j(h_j + \Theta)\right)\right] &= \int_{\mathbb{R}} \phi(z) \prod_{j=1}^{k} \Phi\left(u_j(h_j + \sigma z)\right) \mathrm{d}z \\ &= \Phi_k\left((u_1 h_1, \ldots, u_k h_k)^\top; \boldsymbol{I}_k + \sigma^2 u_{1:k} u_{1:k}^\top\right).\end{aligned}$$

Proof. The first equality follows since Θ has the same distribution as σZ with $Z \sim \mathrm{N}(0, 1)$. For

the second equality, write

$$X_{1:k} := (u_1(h_1 + \Theta), \ldots, u_k(h_k + \Theta))^\top = (u_1 h_1, \ldots, u_k h_k)^\top + u_{1:k}\Theta.$$

Then $X_{1:k} \sim \mathrm{MVN}_k\left((u_1 h_1, \ldots, u_k h_k)^\top, \sigma^2 u_{1:k} u_{1:k}^\top\right)$, and Lemma 1 proves the result. □

Remark 1. *Lemma 2 has the following interpretation. Let $Z_1, \ldots, Z_k$ be conditionally independent given Θ with $Z_j \mid \Theta \sim \mathrm{Bin}(1, \Phi(h_j + \Theta))$. Since $1 - \Phi(a) = \Phi(-a)$ for $a \in \mathbb{R}$, we have $\mathbb{P}(Z_j = z_j \mid \Theta) = \Phi(u_j(h_j + \Theta))$ with $u_j := 2z_j - 1$ for $z_j \in \{0, 1\}$. Hence Lemma 2 gives the joint probability $\mathbb{P}(Z_{1:k} = z_{1:k})$ in closed form. The covariance matrix $\sigma^2 u_{1:k} u_{1:k}^\top$ is of rank one, reflecting the single random effect shared by all trials, and the one-dimensional integral representation is convenient for computation when k is large.*

3. Binomial thinning model for bivariate claim frequency

In this section, we develop a model for two nested claim counts. For each region i and period t, let N_{it} denote the total number of claims and let M_{it} denote the number of extreme claims among them, for example, the claims whose amounts exceed a given threshold. By construction,

$$0 \le M_{it} \le N_{it}.$$

A natural way to respect this constraint is to regard M_{it} as a binomial thinning of N_{it}: each of the N_{it} claims is classified as extreme or not. Repeated observations of the same region over time are linked through two region-specific random effects, one for the claim frequency and one for the probability that a claim is extreme. Throughout this section, the two random effects are assumed to be independent and static over time, and predictions are obtained as exact Bayesian posterior means.

Model 1 (Poisson-gamma binomial-probit thinning model)**.** *Let*

$$r, \sigma \in \mathbb{R}_{>0}, \qquad \alpha, \beta \in \mathbb{R}^p,$$

and let $x_{it} \in \mathbb{R}^p$, $i = 1, \ldots, I$, $t \in \mathbb{N}$, be predetermined exogenous covariates, which are allowed to vary over time. Define

$$\lambda_{it} := \exp\left(x_{it}^\top \alpha\right) \in \mathbb{R}_{>0}, \qquad \eta_{it} := x_{it}^\top \beta \in \mathbb{R}.$$

We define the bivariate sequences $((N_{it}, M_{it}))_{t \in \mathbb{N}}$, $i = 1, \ldots, I$, through the following random effect assumption and observation equations.

i. Random effects: For each region i, assume

$$\Theta_i^{(1)} \sim \mathrm{Gamma}(r, r), \qquad \Theta_i^{(2)} \sim \mathrm{N}(0, \sigma^2),$$

where $\Theta_i^{(1)}$ *and* $\Theta_i^{(2)}$ *are independent. Furthermore, the pairs* $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, $i = 1, \ldots, I$, *are mutually independent.*

ii. Observation equations: Conditional on $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, *assume that the pairs* (N_{it}, M_{it}), $t \in \mathbb{N}$, *are mutually independent with*

$$N_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Pois}\left(\lambda_{it}\Theta_i^{(1)}\right)$$

and

$$M_{it} \mid N_{it}, \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Bin}\left(N_{it}, \Phi\left(\eta_{it} + \Theta_i^{(2)}\right)\right),$$

Furthermore, the regions are independent: the sequences $((N_{it}, M_{it}))_{t\in\mathbb{N}}$, $i = 1, \ldots, I$, *are mutually independent.*

Here, $\text{Bin}(0, p)$ is the point mass at 0, so that $M_{it} = 0$ whenever $N_{it} = 0$.

Since $\mathbb{E}\left[\Theta_i^{(1)}\right] = 1$ and $\mathbb{E}\left[\Theta_i^{(2)}\right] = 0$, the intercepts in α and β are identifiable, and λ_{it} is the a priori expected total claim count. The random effect $\Theta_i^{(1)}$ captures the unobserved heterogeneity of the claim frequency across regions, whereas $\Theta_i^{(2)}$ captures the unobserved heterogeneity of the propensity of a claim to be extreme. For notational convenience, write

$$p_{it}(\theta) := \Phi\left(\eta_{it} + \theta\right), \qquad \theta \in \mathbb{R}.$$

Remark 2. *All results in this section use* λ_{it} *and* η_{it} *only as known constants. Hence they remain valid for any* $\lambda_{it} \in \mathbb{R}_{>0}$ *and* $\eta_{it} \in \mathbb{R}$*; in particular, an exposure offset* $e_{it} > 0$ *can be included through* $\lambda_{it} = e_{it} \exp\left(x_{it}^\top \alpha\right)$.

The following lemma is the well-known thinning property of the Poisson distribution. It shows that, conditional on the random effects, the extreme and non-extreme claim counts are independent Poisson random variables.

Lemma 3 (Poisson thinning)**.** *Under Model 1, conditional on* $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, *the random variables* M_{it} *and* $N_{it} - M_{it}$ *are independent with*

$$M_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Pois}\left(\lambda_{it}\Theta_i^{(1)} p_{it}\left(\Theta_i^{(2)}\right)\right)$$

and

$$N_{it} - M_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Pois}\left(\lambda_{it}\Theta_i^{(1)}\left\{1 - p_{it}\left(\Theta_i^{(2)}\right)\right\}\right).$$

Proof. Fix $\theta_1 > 0$ and $\theta_2 \in \mathbb{R}$, and write $\Lambda := \lambda_{it}\theta_1$ and $p := p_{it}(\theta_2)$. For $m, k \in \mathbb{N}_0$, Model 1 gives

$$\begin{aligned}
&\mathbb{P}\left(M_{it} = m,\, N_{it} - M_{it} = k \mid \Theta_i^{(1)} = \theta_1, \Theta_i^{(2)} = \theta_2\right) \\
&\quad = \mathbb{P}\left(N_{it} = m + k \mid \Theta_i^{(1)} = \theta_1\right) \binom{m+k}{m} p^m (1-p)^k \\
&\quad = \frac{e^{-\Lambda}\Lambda^{m+k}}{(m+k)!} \frac{(m+k)!}{m!\,k!} p^m (1-p)^k \\
&\quad = \frac{e^{-\Lambda p}(\Lambda p)^m}{m!} \cdot \frac{e^{-\Lambda(1-p)}\{\Lambda(1-p)\}^k}{k!},
\end{aligned}$$

which proves the result. □

Since the regions are independent, it suffices to study a single region. The following remark collects the notation for its observed history.

Remark 3. *Fix a region i and $t \in \mathbb{N}$, and consider a realized history*

$$\mathcal{H}_{it} := \{(N_{ij}, M_{ij}) = (n_{ij}, m_{ij}) :\ j = 1, \dots, t\}, \qquad n_{ij}, m_{ij} \in \mathbb{N}_0, \quad m_{ij} \le n_{ij}.$$

Define the cumulative claim count and the cumulative a priori frequency by

$$n_{i\bullet t} := \sum_{j=1}^{t} n_{ij}, \qquad \lambda_{i\bullet t} := \sum_{j=1}^{t} \lambda_{ij}.$$

Each of the $n_{i\bullet t}$ observed claims is regarded as a Bernoulli trial, which is a success if the claim is extreme. To this end, for $j = 1, \dots, t$, define the sign vector

$$u^{(j)} := \big(\underbrace{1, \dots, 1}_{m_{ij}}, \underbrace{-1, \dots, -1}_{n_{ij} - m_{ij}}\big)^\top \in \{-1, 1\}^{n_{ij}},$$

which is empty when $n_{ij} = 0$, and stack these vectors as

$$u_{1:n_{i\bullet t}} := \left(u^{(1)\top}, \dots, u^{(t)\top}\right)^\top, \qquad h_{1:n_{i\bullet t}} := \big(\underbrace{\eta_{i1}, \dots, \eta_{i1}}_{n_{i1}}, \dots, \underbrace{\eta_{it}, \dots, \eta_{it}}_{n_{it}}\big)^\top.$$

That is, u_k is the sign of the kth claim and h_k is the value of η_{ij} in the period j of the kth claim. We then define

$$\mu_{it} := (u_1 h_1, \dots, u_{n_{i\bullet t}} h_{n_{i\bullet t}})^\top \in \mathbb{R}^{n_{i\bullet t}}, \qquad \boldsymbol{U}_{it} := u_{1:n_{i\bullet t}} u_{1:n_{i\bullet t}}^\top \in \mathbb{R}^{n_{i\bullet t} \times n_{i\bullet t}},$$

which are the mean vector and the rank-one matrix appearing in Lemma 2 for the trials formed by the observed claims.

For $n, m \in \mathbb{N}_0$ with $m \le n$, let $\mu_{it}^{(n,m)} \in \mathbb{R}^{n_{i\bullet t}+n}$ and $\boldsymbol{U}_{it}^{(n,m)} \in \mathbb{R}^{(n_{i\bullet t}+n)\times(n_{i\bullet t}+n)}$ denote the corresponding quantities for the augmented history $\mathcal{H}_{it} \cup \{(N_{i,t+1}, M_{i,t+1}) = (n, m)\}$. That is, n

additional trials with η-value $\eta_{i,t+1}$ are appended, m of them with sign 1 and $n-m$ of them with sign -1.

Finally, define

$$D_{it}(\theta) := \prod_{j=1}^{t} p_{ij}(\theta)^{m_{ij}} \{1 - p_{ij}(\theta)\}^{n_{ij}-m_{ij}}, \qquad \theta \in \mathbb{R}.$$

Since $1 - \Phi(a) = \Phi(-a)$ for $a \in \mathbb{R}$, we have

$$D_{it}(\theta) = \prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k(h_k + \theta)\right). \tag{3.1}$$

For $n_{i\bullet t} = 0$, we adopt the conventions $D_{it}(\theta) = 1$ and $\Phi_0\left(\mu_{it}; \boldsymbol{I}_0 + \sigma^2 \boldsymbol{U}_{it}\right) = 1$.

The following theorem shows that the joint probability of the observed history factorizes into a closed-form Poisson-gamma part, which depends only on (α, r), and a Bernoulli-probit part of the form in Lemma 2, which depends only on (β, σ).

Theorem 1 (Likelihood factorization)**.** *Consider Model 1 with the realized history $\mathcal{H}_{it}$, and let the notation be as in Remark 3. Then*

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = L_{it}^{(1)} L_{it}^{(2)}, \tag{3.2}$$

where

$$L_{it}^{(1)} := \mathbb{P}\left(N_{i,1:t} = n_{i,1:t}\right) = \frac{\Gamma\left(r + n_{i\bullet t}\right)}{\Gamma(r) \prod_{j=1}^{t} n_{ij}!} \frac{r^r \prod_{j=1}^{t} \lambda_{ij}^{n_{ij}}}{\left(r + \lambda_{i\bullet t}\right)^{r + n_{i\bullet t}}} \tag{3.3}$$

and

$$\begin{aligned} L_{it}^{(2)} &:= \mathbb{P}\left(M_{i,1:t} = m_{i,1:t} \mid N_{i,1:t} = n_{i,1:t}\right) \\ &= \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} \int_{\mathbb{R}} \frac{1}{\sigma} \phi\left(\frac{\theta}{\sigma}\right) D_{it}(\theta)\, \mathrm{d}\theta \\ &= \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} \Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right). \end{aligned} \tag{3.4}$$

Proof. By the conditional independence in Model 1,

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)}\right) \mathbb{P}\left(M_{ij} = m_{ij} \mid N_{ij} = n_{ij}, \Theta_i^{(2)}\right)\right].$$

The first factor in the product depends only on $\Theta_i^{(1)}$ and the second only on $\Theta_i^{(2)}$. Since $\Theta_i^{(1)}$ and $\Theta_i^{(2)}$ are independent, we obtain

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)}\right)\right] \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(M_{ij} = m_{ij} \mid N_{ij} = n_{ij}, \Theta_i^{(2)}\right)\right].$$

For the first expectation, the gamma integral gives

$$\mathbb{E}\left[\prod_{j=1}^{t}\mathbb{P}\left(N_{ij}=n_{ij}\mid\Theta_i^{(1)}\right)\right]=\int_0^\infty\prod_{j=1}^{t}\frac{e^{-\lambda_{ij}\theta}(\lambda_{ij}\theta)^{n_{ij}}}{n_{ij}!}\,\frac{r^r}{\Gamma(r)}\theta^{r-1}e^{-r\theta}\,\mathrm{d}\theta$$
$$=\frac{r^r\prod_{j=1}^{t}\lambda_{ij}^{n_{ij}}}{\Gamma(r)\prod_{j=1}^{t}n_{ij}!}\int_0^\infty\theta^{r+n_{i\bullet t}-1}e^{-(r+\lambda_{i\bullet t})\theta}\,\mathrm{d}\theta,$$

which equals the right-hand side of (3.3). Summing the first display over all $m_{i,1:t}$ with $0\le m_{ij}\le n_{ij}$ shows that this expectation is $\mathbb{P}\left(N_{i,1:t}=n_{i,1:t}\right)$, because the second expectation sums to one.

For the second expectation, we have

$$\mathbb{E}\left[\prod_{j=1}^{t}\mathbb{P}\left(M_{ij}=m_{ij}\mid N_{ij}=n_{ij},\Theta_i^{(2)}\right)\right]=\prod_{j=1}^{t}\binom{n_{ij}}{m_{ij}}\mathbb{E}\left[D_{it}\left(\Theta_i^{(2)}\right)\right],$$

which proves the second line of (3.4). It then follows from (3.2) and (3.3) that this expression is $\mathbb{P}\left(M_{i,1:t}=m_{i,1:t}\mid N_{i,1:t}=n_{i,1:t}\right)$. Finally, by (3.1),

$$\mathbb{E}\left[D_{it}\left(\Theta_i^{(2)}\right)\right]=\mathbb{E}\left[\prod_{k=1}^{n_{i\bullet t}}\Phi\left(u_k\left(h_k+\Theta_i^{(2)}\right)\right)\right],$$

and applying Lemma 2 with $k=n_{i\bullet t}$, $u_{1:k}$, and $h_{1:k}$ proves the last line of (3.4). □

Remark 4 (Separable estimation)**.** *By Theorem 1 and the independence of the regions, the log-likelihood of the observations $\mathcal{H}_{1T},\ldots,\mathcal{H}_{IT}$ decomposes as*

$$\ell(\alpha,r,\beta,\sigma)=\ell^{(1)}(\alpha,r)+\ell^{(2)}(\beta,\sigma),\qquad\ell^{(k)}:=\sum_{i=1}^{I}\log L_{iT}^{(k)},\quad k=1,2.$$

Hence the maximum likelihood estimators of (α,r) and (β,σ) are obtained from two separate optimizations, and the Fisher information matrix is block-diagonal. Here, $\ell^{(1)}$ is the log-likelihood of a negative binomial random effect panel model for N_{it}, and $\ell^{(2)}$ is the log-likelihood of a binomial probit model for M_{it} given N_{it} with a Gaussian random intercept. Periods with $n_{it}=0$ do not contribute to $\ell^{(2)}$.

Remark 5 (Computation)**.** *The dimension $n_{i\bullet t}$ of the multivariate normal distribution function in (3.4) equals the cumulative number of claims of region i, which can be large. However, since $\boldsymbol{U}_{it}$ is of rank one, the second line of (3.4) expresses the same quantity as the one-dimensional integral*

$$\Phi_{n_{i\bullet t}}\left(\mu_{it};\boldsymbol{I}_{n_{i\bullet t}}+\sigma^2\boldsymbol{U}_{it}\right)=\int_{\mathbb{R}}\phi(z)\,D_{it}(\sigma z)\,\mathrm{d}z,$$

which depends on the data only through the t pairs (n_{ij},m_{ij}) and can be evaluated accurately by Gauss–Hermite quadrature.

The following corollary shows that the two random effects remain independent a posteriori, and that the frequency random effect has a conjugate gamma posterior.

Corollary 1 (Posterior distribution of the random effects)**.** *Under the setting of Theorem 1, the following results hold.*

i. Conditional on $\mathcal{H}_{it}$, the random effects $\Theta_i^{(1)}$ and $\Theta_i^{(2)}$ are independent.

ii. We have

$$\Theta_i^{(1)} \mid \mathcal{H}_{it} \sim \text{Gamma}\left(r + n_{i\bullet t},\, r + \lambda_{i\bullet t}\right).$$

iii. The conditional distribution $\Theta_i^{(2)} \mid \mathcal{H}_{it}$ has density $\pi(\theta \mid \mathcal{H}_{it})$ at $\theta \in \mathbb{R}$ given by

$$\pi(\theta \mid \mathcal{H}_{it}) = \frac{\frac{1}{\sigma}\phi\left(\frac{\theta}{\sigma}\right)\prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k(h_k+\theta)\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}. \tag{3.5}$$

Proof. Let π_1 and π_2 denote the prior densities of $\Theta_i^{(1)}$ and $\Theta_i^{(2)}$, respectively. By Bayes' formula and the proof of Theorem 1, the joint posterior density of $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$ at $(\theta_1, \theta_2) \in \mathbb{R}_{>0} \times \mathbb{R}$ is

$$\frac{\pi_1(\theta_1)\prod_{j=1}^t \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)} = \theta_1\right)}{L_{it}^{(1)}} \cdot \frac{\pi_2(\theta_2)\prod_{j=1}^t \binom{n_{ij}}{m_{ij}} D_{it}(\theta_2)}{L_{it}^{(2)}}.$$

Each factor integrates to one, which proves part i. The first factor is proportional to

$$\theta_1^{r+n_{i\bullet t}-1} e^{-(r+\lambda_{i\bullet t})\theta_1},$$

which proves part ii. Substituting (3.1) and (3.4) into the second factor proves part iii. □

By part ii of Corollary 1, the posterior of the frequency random effect depends on the history only through $n_{i\bullet t}$; in particular, the extreme claim counts $m_{i,1:t}$ carry no information on $\Theta_i^{(1)}$. Conversely, by part iii, the total claim counts affect the posterior of $\Theta_i^{(2)}$ only by determining the number of Bernoulli trials. The density (3.5) is a Gaussian density multiplied by a product of probit functions, normalized by the multivariate normal distribution function in Lemma 2.

We now present the exact Bayesian one-step-ahead predictions.

Theorem 2 (Closed-form one-step prediction)**.** *Consider Model 1 with the realized history $\mathcal{H}_{it}$, and let the notation be as in Remark 3. Then, we have the following results.*

i. For $n, m \in \mathbb{N}_0$ with $m \leq n$, the predictive probability is

$$\begin{aligned}
&\mathbb{P}\left(N_{i,t+1} = n,\, M_{i,t+1} = m \mid \mathcal{H}_{it}\right) \\
&\quad = \frac{\Gamma\left(r + n_{i\bullet t} + n\right)}{\Gamma\left(r + n_{i\bullet t}\right) n!} \left(\frac{r + \lambda_{i\bullet t}}{r + \lambda_{i\bullet t} + \lambda_{i,t+1}}\right)^{r+n_{i\bullet t}} \left(\frac{\lambda_{i,t+1}}{r + \lambda_{i\bullet t} + \lambda_{i,t+1}}\right)^{n} \\
&\qquad \times \binom{n}{m} \frac{\Phi_{n_{i\bullet t}+n}\left(\mu_{it}^{(n,m)}; \boldsymbol{I}_{n_{i\bullet t}+n} + \sigma^2 \boldsymbol{U}_{it}^{(n,m)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.
\end{aligned}$$

ii. The one-step-ahead prediction of the total claim count is

$$\mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}} = \lambda_{i,t+1} \left\{\zeta_{it} \frac{n_{i\bullet t}}{\lambda_{i\bullet t}} + (1 - \zeta_{it})\right\}, \qquad \zeta_{it} := \frac{\lambda_{i\bullet t}}{r + \lambda_{i\bullet t}}.$$

iii. The predictive probability that a claim in period $t+1$ is extreme is

$$\widehat{p}_{i,t+1} := \mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right) \mid \mathcal{H}_{it}\right] = \frac{\Phi_{n_{i\bullet t}+1}\left(\mu_{it}^{(1,1)}; \boldsymbol{I}_{n_{i\bullet t}+1} + \sigma^2 \boldsymbol{U}_{it}^{(1,1)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.$$

iv. For $n \in \mathbb{N}_0$, the one-step-ahead prediction of the extreme claim count given the total claim count is

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\, \widehat{p}_{i,t+1}.$$

v. The one-step-ahead prediction of the extreme claim count is

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}} \widehat{p}_{i,t+1}.$$

Proof. We first show part i. Applying Theorem 1 to the augmented history $\mathcal{H}_{it} \cup \{(N_{i,t+1}, M_{i,t+1}) = (n, m)\}$ and to $\mathcal{H}_{it}$, we obtain

$$\mathbb{P}\left(N_{i,t+1} = n,\, M_{i,t+1} = m \mid \mathcal{H}_{it}\right) = \frac{L_{i,t+1}^{(1)}}{L_{it}^{(1)}} \cdot \frac{L_{i,t+1}^{(2)}}{L_{it}^{(2)}},$$

where $L_{i,t+1}^{(1)}$ and $L_{i,t+1}^{(2)}$ are evaluated at the augmented history. By (3.3),

$$\frac{L_{i,t+1}^{(1)}}{L_{it}^{(1)}} = \frac{\Gamma\left(r + n_{i\bullet t} + n\right)}{\Gamma\left(r + n_{i\bullet t}\right) n!} \frac{\lambda_{i,t+1}^{n} \left(r + \lambda_{i\bullet t}\right)^{r+n_{i\bullet t}}}{\left(r + \lambda_{i\bullet t} + \lambda_{i,t+1}\right)^{r+n_{i\bullet t}+n}},$$

and, by the last line of (3.4),

$$\frac{L_{i,t+1}^{(2)}}{L_{it}^{(2)}} = \binom{n}{m} \frac{\Phi_{n_{i\bullet t}+n}\left(\mu_{it}^{(n,m)}; \boldsymbol{I}_{n_{i\bullet t}+n} + \sigma^2 \boldsymbol{U}_{it}^{(n,m)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)},$$

which proves part i.

To show part ii, note that, conditional on $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, $N_{i,t+1}$ is independent of $\mathcal{H}_{it}$. Hence, by part ii of Corollary 1,

$$\mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1}\mathbb{E}\left[\Theta_i^{(1)} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1}\frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}},$$

and the second expression follows from $(r + n_{i\bullet t})/(r + \lambda_{i\bullet t}) = \zeta_{it}\, n_{i\bullet t}/\lambda_{i\bullet t} + 1 - \zeta_{it}$.

To show part iii, part iii of Corollary 1 gives

$$\widehat{p}_{i,t+1} = \frac{\mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right)\prod_{k=1}^{n_{i\bullet t}}\Phi\left(u_k\left(h_k + \Theta_i^{(2)}\right)\right)\right]}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.$$

Since $p_{i,t+1}(\theta) = \Phi\left(1 \cdot (\eta_{i,t+1} + \theta)\right)$, the product in the numerator is the product in (3.1) for the history augmented by $(n, m) = (1, 1)$. Applying Lemma 2 to the augmented history proves part iii.

To show part iv, we first observe that, conditional on $\mathcal{H}_{it}$, the joint density of $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$ and the probability mass function of $N_{i,t+1}$ satisfy

$$\pi(\theta_1, \theta_2, n \mid \mathcal{H}_{it}) = \pi(\theta_1 \mid \mathcal{H}_{it})\mathbb{P}\left(N_{i,t+1} = n \mid \Theta_i^{(1)} = \theta_1\right)\pi(\theta_2 \mid \mathcal{H}_{it}),$$

by part i of Corollary 1 and Model 1. Hence $\Theta_i^{(2)}$ is independent of $N_{i,t+1}$ conditional on $\mathcal{H}_{it}$. Together with the binomial observation equation, this gives

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\,\mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right) \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\,\widehat{p}_{i,t+1}.$$

Finally, part v follows from part iv and part ii, since

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}\right] = \mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right]\widehat{p}_{i,t+1}.$$

□

Remark 6 (Interpretation of the predictions)**.** *Part ii of Theorem 2 is the exact credibility formula of the Poisson-gamma model: the a priori frequency $\lambda_{i,t+1}$ is multiplied by a credibility-weighted average of the observed ratio $n_{i\bullet t}/\lambda_{i\bullet t}$ and the prior mean 1, with credibility factor ζ_{it}, which increases to one as $\lambda_{i\bullet t} \to \infty$. Part iii is the posterior predictive probability that an additional Bernoulli-probit trial, namely a new claim, is a success. In particular, if $n_{i\bullet t} = 0$, then $\widehat{p}_{i,t+1} = \Phi\left(\eta_{i,t+1}/\sqrt{1+\sigma^2}\right)$, which is the a priori probability that a claim is extreme. Part v shows that the prediction of the extreme claim count is the product of the two exact Bayesian predictions, one for the frequency and one for the proportion of extreme claims, as a consequence of the a posteriori independence of the random effects.*

We conclude this section with the unconditional moments of Model 1, which describe the dependence induced by the random effects.

Proposition 1 (Moments)**.** *Under Model 1, fix a region i, and define*

$$\pi_{it} := \Phi\left(\frac{\eta_{it}}{\sqrt{1+\sigma^2}}\right), \qquad \pi_{i,st} := \Phi_2\left((\eta_{is},\eta_{it})^\top; \boldsymbol{I}_2 + \sigma^2 \boldsymbol{J}_2\right), \qquad s,t \in \mathbb{N},$$

where $\boldsymbol{J}_2$ is the 2×2 matrix of ones. Then, for $s,t \in \mathbb{N}$ with $s \neq t$, we have

$$\begin{aligned}
\mathbb{E}\left[N_{it}\right] &= \lambda_{it}, & \mathbb{E}\left[M_{it}\right] &= \lambda_{it}\pi_{it},\\
\mathrm{Var}\left(N_{it}\right) &= \lambda_{it} + \frac{\lambda_{it}^2}{r}, & \mathrm{Cov}\left(N_{is},N_{it}\right) &= \frac{\lambda_{is}\lambda_{it}}{r},\\
\mathrm{Cov}\left(N_{it},M_{it}\right) &= \pi_{it}\left(\lambda_{it} + \frac{\lambda_{it}^2}{r}\right), & \mathrm{Cov}\left(N_{is},M_{it}\right) &= \frac{\lambda_{is}\lambda_{it}\pi_{it}}{r},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{Var}\left(M_{it}\right) &= \lambda_{it}\pi_{it} + \lambda_{it}^2\left\{\left(1+\frac{1}{r}\right)\pi_{i,tt} - \pi_{it}^2\right\},\\
\mathrm{Cov}\left(M_{is},M_{it}\right) &= \lambda_{is}\lambda_{it}\left\{\left(1+\frac{1}{r}\right)\pi_{i,st} - \pi_{is}\pi_{it}\right\}.
\end{aligned}$$

Proof. Write $\Theta_1 := \Theta_i^{(1)}$ and $\Theta_2 := \Theta_i^{(2)}$, and recall $\mathbb{E}\left[\Theta_1\right] = 1$ and $\mathbb{E}\left[\Theta_1^2\right] = 1 + 1/r$. By Lemma 2 with $k=1$ and $k=2$, and $u_{1:k}$ the vector of ones,

$$\mathbb{E}\left[p_{it}(\Theta_2)\right] = \pi_{it}, \qquad \mathbb{E}\left[p_{is}(\Theta_2)p_{it}(\Theta_2)\right] = \pi_{i,st}, \qquad s,t \in \mathbb{N}.$$

Conditional on (Θ_1,Θ_2), Model 1 and Lemma 3 give, for $s \neq t$,

$$\begin{aligned}
\mathbb{E}\left[N_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1, & \mathbb{E}\left[M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 p_{it}(\Theta_2),\\
\mathbb{E}\left[N_{it}^2 \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 + \lambda_{it}^2\Theta_1^2, & \mathbb{E}\left[M_{it}^2 \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 p_{it}(\Theta_2) + \lambda_{it}^2\Theta_1^2 p_{it}(\Theta_2)^2,
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}\left[N_{it}M_{it} \mid \Theta_1,\Theta_2\right] &= p_{it}(\Theta_2)\mathbb{E}\left[N_{it}^2 \mid \Theta_1,\Theta_2\right] = p_{it}(\Theta_2)\left(\lambda_{it}\Theta_1 + \lambda_{it}^2\Theta_1^2\right),\\
\mathbb{E}\left[N_{is}M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{is}\lambda_{it}\Theta_1^2 p_{it}(\Theta_2),\\
\mathbb{E}\left[M_{is}M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{is}\lambda_{it}\Theta_1^2 p_{is}(\Theta_2)p_{it}(\Theta_2).
\end{aligned}$$

Taking expectations, using the independence of Θ_1 and Θ_2, and subtracting the products of the means proves the result. □